

App Development Roadmap (Beginner → Advanced)

1. Learn Programming Basics

Start with one language first.

Best Languages for App Development

- **JavaScript** → Best for web + mobile apps
- **Python** → Easy for beginners
- **Dart** → Used in Flutter
- **Kotlin** → Android apps
- **Swift** → iPhone apps

First Topics

- Variables
- Loops
- Functions
- Arrays
- OOP Basics
- File handling

Platforms

- [freeCodeCamp](#)
 - [W3Schools](#)
 - [MDN Web Docs](#)
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2. Learn Web Development Basics

Even mobile developers should know this.

Frontend

Learn:

- HTML
- CSS
- JavaScript

Build Projects

- Portfolio Website
- Calculator
- To-Do App
- Weather App

Framework

Learn:

- [React](#)
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3. Choose Your App Development Path

OPTION A — Android Development

Learn

- Java or Kotlin
- Android Studio
- XML UI
- APIs
- Firebase

Official Tools

- [Android Studio](#)
- [Kotlin](#)

Projects

- Notes App

- Chat App
 - Expense Tracker
 - Camera App
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OPTION B — Cross Platform (BEST for Beginners)

Learn Flutter

One code → Android + iPhone + Web

Learn

- Dart
- Flutter Widgets
- State Management
- Firebase
- REST APIs

Official Sites

- [Flutter](#)
- [Dart](#)

Projects

- WhatsApp Clone
 - Food Delivery App
 - AI Chat App
 - IoT Dashboard
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OPTION C — React Native

Learn

- JavaScript
- React
- React Native

Official Sites

- [React Native](#)
 - [Expo](#)
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4. Learn Backend Development

Topics

- APIs
- Authentication
- Databases
- Cloud Storage
- Real-time Communication

Backend Technologies

- Node.js
- Express.js
- MongoDB
- Firebase
- PostgreSQL

Official Sites

- [Node.js](#)
 - [MongoDB](#)
 - [Firebase](#)
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5. Learn Git & GitHub

Learn

- git init
- commit
- push
- branches
- pull requests

Platform

- [GitHub](#)

Build:

- Strong GitHub Profile
 - Project Documentation
 - README files
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6. Intermediate Level Skills

Learn

- Authentication
- JWT
- API Integration
- Local Storage
- Notifications
- Maps & GPS
- Payment Gateway
- Socket Programming

Build

- Uber Clone
 - Instagram Clone
 - E-commerce App
 - Video Calling App
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7. Advanced Level

Learn

- Clean Architecture
- Design Patterns
- CI/CD
- Docker
- Kubernetes
- Microservices
- WebSockets
- App Security
- Performance Optimization

Tools

- [Docker](#)
 - [Kubernetes](#)
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8. AI + App Development (Power Combo)

Since AI apps are growing fast, learn:

- AI APIs
- Chatbots
- Voice Assistants
- Computer Vision
- Edge AI

Tools

- [OpenAI Platform](#)
- [TensorFlow](#)
- [PyTorch](#)

Projects

- AI Study Assistant
- Smart CCTV

- Face Detection App
 - AI Farming App
 - Drone Monitoring App
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9. Embedded + IoT App Development

This matches your Raspberry Pi + LoRa interest.

Learn

- MQTT
- ESP32
- Raspberry Pi
- Sensor Data Apps
- Bluetooth/WiFi Communication

Tools

- [Raspberry Pi](#)
- [Arduino](#)

Projects

- Smart Home App
 - Drone Controller
 - Animal Tracking System
 - LoRa Communication Dashboard
 - Smart Agriculture System
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10. Deployment & Publishing

Android

- Google Play Store

iPhone

- Apple App Store

Web Apps

- [Vercel](#)
 - [Netlify](#)
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BEST LEARNING ORDER

Month 1–2

- Programming Basics
- HTML/CSS/JS

Month 3–4

- React or Flutter
- Small Projects

Month 5–6

- Backend + Database
- APIs

Month 7–8

- Full Stack Apps
- Firebase
- Authentication

Month 9–12

- Advanced Apps
- AI Integration
- IoT + Embedded Apps

- Deploy Real Apps
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Best YouTube Channels

Web & App Dev

- Hitesh Choudhary
 - CodeWithHarry
 - Traversy Media
 - freeCodeCamp
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Final Goal

By the advanced stage, you should be able to build:

- AI Apps
- SaaS Products
- Startup MVPs
- IoT Control Apps
- Embedded Dashboards
- Real-time Systems
- Drone Control Interfaces
- Defense/Surveillance Apps